

Cipher Decryption Walkthrough Report

1. Recovered Plaintext

BREAKER OF THIS CODE WILL BE BLESSED BY THE SQUEAKY SPIRIT RESIDING IN THE HOLE GO AHEAD AND FIND A WAY OF BREAKING THE SPELL ON HIM CAST BY THE EVIL JAFFAR THE SPIRIT OF THE CAVE MAN IS ALWAYS WITH YOU FIND THE MAGIC WAND THAT WILL LET YOU OUT OF THE CAVES IT WOULD MAKE YOU A MAGICIAN NO LESS THAN JAFFAR SPEAK THE PASSWORD THE MAGIC OF WAND TO GO THROUGH

2. Encryption Algorithm / Cipher

Two-layer encryption scheme consisting of: (a) a block permutation/transposition cipher with block size 5 and permutation order [3, 2, 4, 0, 1], followed by (b) a monoalphabetic substitution cipher.

3. Encryption Keys

Permutation Key (block size 5): [3, 2, 4, 0, 1]

a→q, b→j, c→e, d→p, e→v, f→s, g→g, h→f, i→c, j→k, k→m, l→t, m→u, n→y, o→w, p→h, q→i, r→n, s→l, t→a, u→d, v→b, w→r, x→o, y→x, z→z

4. Approach for Problem Solution

Step 1: Remove all spaces from the ciphertext to obtain a continuous encrypted stream.

Step 2: Apply the inverse block permutation using block size 5 and order [3,2,4,0,1].

Step 3: Construct the inverse substitution mapping from the provided substitution key.

Step 4: Replace each character in the permuted text using the inverse substitution table.

Step 5: Convert the result to uppercase and insert word spacing to recover the final plaintext.

Step 6: Verify that the recovered message forms meaningful English text, confirming successful decryption.